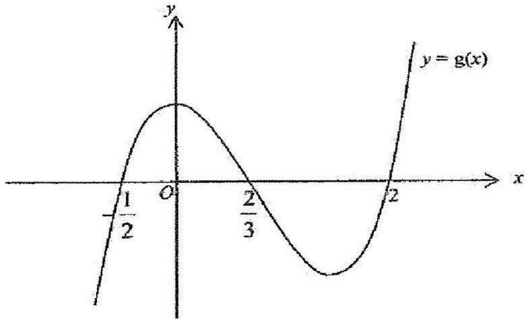
	Unit 4 Polynomials, Remainder and Factor Theorem
1.	(a) The cubic polynomial $f(x)$ is such that the roots of $f(x) = 0$ are $x = 2, x =- 1$ and $x = k$ and the coefficient of x^3 is 1 . Calculate the value of k if $f(x)$ leaves a remainder of -128 when divided by $(x - 3)$. (b) Given that $2x^3 - 4x^2 - px - 2 \equiv (x - 1)(x + 2)Q(x) + 5x + q$, where $Q(x)$ is the quotient. (i) State the degree of $Q(x)$. (ii) Find the value of p and of q. (c) A sum of money $\$\left(3x^3 - 5x^2 + 6x - 54\right)$ is to be divided equally among $(x - 3)$ grandchildren. Find the share of money, in terms of x, each grandchild will receive. Ans: (a) 35 (b)(i) 1 (ii) $p = 15; q =- 14$ (c) $\$\left(3x^2 + 4x + 18\right)$
2.	(a) Given that $f(x) = ax^3 + bx^2 + 2$ leaves a remainder of -2 when divided by $(x + 2)$ and $3f(x) + 6$ is exactly divisible by $(x - 1)$. Find the value of a and of b. (b) (i) Factorise $6x^3 - 7x^2 - 8x + 5$ completely. (ii) Hence, solve the equation $5x^3 - 8x^2 - 7x + 6 = 0$. Ans: (a) $a =- 1, b =- 3$ (b)(i) $(x + 1)(2x - 1)(3x - 5)$ (ii) $x =- 1, x = 2, x = \frac{3}{5}$
3.	The term containing the highest power of x in the polynomial $f(x)$ is $2x^3$. Two of the roots of the equation $f(x) = 0$ are -4 and 2 . It is given that $f(x)$ leaves a remainder of 5 when divided by $x + 3$. (i) Express $f(x)$ in descending powers of x. (ii) Find the remainder when $f(x)$ is divided by $2x - 1$. Ans: (i) $f(x) = 2x^3 + 9x^2 - 6x - 40$ (ii) -40
4.	The cubic polynomial $f(x)$ is such that the coefficient of x^3 is 3 and its constant term is -8 . The roots of the equation $f(x) = 0$ are $-\frac{1}{3}$, p and q. Given that $f(x)$ has a remainder of -20 when divided by $x - 1$ and p is a positive integer, find (i) the value of p and of q, (ii) the reminder when $f(x)$ is divided by $2x - 1$.

	Ans: (i) $p = 2, q =- 4$ (ii) $- 16\frac{7}{8}$
5.	(i) Find the value of h for which $x^3 + hx^2 + 4hx - (3 + h)$ is divisible by $x + 1$. (ii) Using the value of h found in pant (i), solve the equation $x^3 + hx^2 + 4hx - (3 + h) = 0$, expressing non-integer roots in the form $c \pm \sqrt{d}$, where c and d are integers. Ans: (i) -1 (ii) -1 or $1\pm\sqrt{3}$
6.	(i) The expression $f(x) = abx^3 + (a - 2b)x^2 - 4ax - (a + b)$, where $a\neq0$ and $b\neq0$, is exactly divisible by $x^2 - 4$. Find the value of a and b. (ii) Hence factorise $f(x)$ completely. (iii) Find the remainder when $f(x)$ is divided by $(2x - 1)$. Ans: (i) $a = 3, b = 1$ (ii) $f(x) = (x + 2)(x - 2)(3x + 1)$ (iii) Remainder $=- 9\frac{3}{8}$
7.	The expression $px^3 - x^2 + qx + r$, where $p > 0$, leaves a remainder of -14 when divided by $x + 2$, and has a factor of $x^2 + x$. Find the values of p, q and r. Ans: $p = 2, q =- 3, r = 0$
8.	The function f is defined by $f(x) = 2x^3 + 4ax^2 - 6x^2 + (10 - 15a)x + 11a - 6$, where a is a constant. (i) Show that $x - 1$ is a factor of $f(x)$. (ii) Find the possible values of a for which the equation $f(x) = 0$ only has exactly 2 real roots. Ans: (ii) $a =- 4, a = 0.5$
9.	The expression $f(x) = 5x^3 + ax^2 + b$ where a and b are constants, has a factor of $x + 1$ and leaves a remainder of 51 when divided by $x - 2$. (i) Find the value of a and of b. (ii) Using the values of a and b found in part (i) determine the number of real roots of the equation $5x^3 + ax^2 + b = 0$, justifying your answer. Ans: (i) $a = 2, b = 3$ (ii) one real root

10.	The term containing the highest power of x in the polynomial $f(x)$ is $2x^4$ and the roots of $f(x) = 0$ are 2 and -7. $f(x)$ has a remainder of -72 when divided by $(x + 1)$, and a remainder of -80 when divided by $(x - 1)$. (i) Find the expression for $f(x)$ in descending power of x. (ii) Explain why the equation $f(x) = 0$ has exactly 2 real roots. Ans: (i) $f(x) = 2x^4 + 13x^3 - 8x^2 - 17x - 70$
11.	The expression $2x^3 + ax^2 + bx - 9$, where a and b are constants, has a factor $x - 3$ and leaves a remainder of 8 when divided by $x + 1$. (i) Find the value of a and of b. (ii) Using the values of a and b found in part (i), factorise the expression $2x^3 + ax^2 + bx - 9$ completely. Hence solve the equation $16x^3 + 4ax^2 = 9 - 2bx$. Ans: (i) $a = 1, b =- 18$ (ii) $f(x) = (x - 3)(2x + 1)(x + 3); x = \frac{3}{2}, -\frac{1}{4}, -\frac{3}{2}$
12.	Given that $x + 5$ is a common factor of $x^3 + kx^2 - Ax + 15$ and $x^3 - x^2 - (A + 5)x + 40$, find the value of k and of A. Hence factorise $x^3 + kx^2 - Ax + 15$ completely. Ans: $A = 17, k = 1; (x + 5)(x - 3)(x - 1)$
13.	It is given that $f(x) = x^3 + ax^2 + bx + 14$ leaves a remainder of 5 when divided by $x^2 - 9$. (i) Find the values of a and of b. (ii) Using your values of a and b, solve the equation $f(x) = 0$. Ans: (i) $a =- 1, b =- 9$ (ii) $2, 2.19$ or -3.19
14.	The polynomial $P(x) = 2x^3 + ax^2 + bx + 8$, where a and b are constants, leaves a remainder of 10 when divided by $2x - 1$. Given that $x + 2$ is a factor of $P(x)$. (i) Find the value of a and of b (ii) Explain why the equation $P(x) = 0$ has only 1 real root. Hence find this root.

	Ans: (i) $a = 3, b = 2$ (ii) $x =- 2$
15.	(a) (i) Show that $(2x + 1)$ is a factor of $f(x)$ where $f(x) = 6x^3 - 13x^2 + 4$ *completely*. (ii) Given that $6x^3 - 13x^2 + 4 = (2x + 1)(ax^2 + bx + c)$, state the values of a, b and c. Factorize $6x^3 - 13x^2 + 4$ completely. (iii) The diagram shows the graph of $y = g(x)$ where $g(x) \neq f(x)$.  Write down a possible equation for the curve. (b) The polynomial $p(x)$ has a factor $(x - 3)$ and when divided by $(x + 1)$ leaves a remainder of 8. Given that $p(x) = (x + 1)(x - 3)q(x) + mx + c$, form two equations in m and c. Hence, find the remainder when $p(x)$ is divided by $(x^2 - 2x - 3)$. Ans: (a)(ii) $a = 3, b =- 8, c = 4, 6x^3 - 13x^2 + 4 = (2x + 1)(x - 2)(3x - 2)$ (iii) $m > 0, m \neq 1$ (b) $- 2x + 10$
16.	The expression $2x^3 + ax^2 + bx + 8$, where a and b are constants, has a factor of $x + 2$ and leaves a remainder of -24 when divided by $x - 2$. (i) Find the values of a and of b. (ii) Using the values of a and of b found in part (i), solve the equation $2x^3 + ax^2 + bx + 8 = 0$. Ans: (i) $a =- 5, b =- 14$ (ii) -2 or $\frac{1}{2}$ or 4
17.	The term containing the highest power of x in the polynomial $f(x)$ is $- 2x^4$. Two of the roots of the equation $P(x) = 0$ are -1 and 2 . The third factor of $f(x)$ is $(x^2 + mx - m)$.

	(a) Write down, without expanding, an expression for the polynomial $f(x)$. (b) In the case where all the roots of the equation $f(x) = 0$ are real, find the range of values of m. (c) In the case where $m =- 4$, find the number of distinct real roots of the equation $f(x) = 0$. (d) In the case where $m =- 1$, explain using your answer in (b) or otherwise, why $\left(x^2 + mx - m\right)$ is always positive. Hence solve the inequality $f(x) > 0$. Ans: (a) $- 2(x + 1)(x - 2)\left(x^2 + mx - m\right)$ (b) $m\leq - 4$ or $m\geq0$ (c) 2(d) $- 1 < x < 2$
18.	$f(x) = 6x^3 + ax^2 + bx - 6$ has a factor $x + 2$ but leaves a remainder of -12 when divided by $x - 1$. (i) Find the value of a and of b. (ii) Factorise $f(x)$ completely and hence solve the equation $48x^3 + 4ax^2 = 6 - 2bx$. Ans: (i) $a = 5, b =- 17$ (ii) $f(x) = (x + 2)(3x - 1)(2x - 3), x =- 1, -\frac{1}{6}, \frac{3}{4}$
19.	The expression $px^3 - 3qx^2 + qx + 2p$ is exactly divisible by $(2x + 1)$ and has a remainder of -10 when divided by $(x + 1)$. (a) Show that the values of p and q are 2 and 3 respectively. (b) Hence, or otherwise, (i) factorise the above expression completely, and (ii) determine the remainder when the expression above is divided by $(x - 3)$. Ans: (b)(i) $(2x + 1)(x - 4)(x - 1)$ (ii) -14
20.	(i) Given that $f(x) = 2x^3 + ax^2 + bx - 30$ has a factor $(x + 3)$ and leaves a reminder of -28 when divided by $(x - 1)$. Find the values of a and of b and solve $f(x) = 0$. (ii) Hence solve $2(y + 1)^3 + a(y + 1)^2 + by + b - 30 = 0$. Ans: $a = 7, b =- 7; x =- 3, 2$ or -2.5 (ii) $y =- 4, 1$ or -3.5
21.	Given the cubic expression $f(x) = x^3 + px^2 + qx + 4$ has a factor $(x + 2)$ and leaves a remainder of 6 when divided by $(x + 1)$, (i) find the value of p and of q,